Tran The Duong

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M.S. student in Electrical Engineering at KAIST, focusing on multi-object tracking and robotic control. Builds perception models, control algorithms, and simulation software in Python, C++, and MATLAB.

Education

KAIST
M.S. in Electrical Engineering | Daejeon, South Korea Sep. 2024 – Present
Expected graduation: Feb. 2027. Advisor: Prof. Dong Eui Chang.

Hanoi University of Science and Technology
B.S. in Automatic Control | Hanoi, Vietnam Sep. 2019 – May 2024
GPA: **3.89/4.00**. Thesis advisor: Assoc. Prof. Dao Phuong Nam.

Research Experience

Graduate Researcher, CtrlLab, KAIST
Computer vision, robotics, and control Sep. 2024 – Present
Current focus: Multi-object tracking (Oct. 2025 – Present).

BeliefCO *Master's thesis, ongoing*
End-to-End Transformer-Based Multi-Object Tracking for Drone and Bird Surveillance

- Developed a Transformer-based tracker that combines motion uncertainty, object-existence estimates, and probabilistic association to maintain object identities.
- Created WOSDETC tracking annotations and recording-level data splits; evaluated the tracker on four aerial-tracking benchmarks.
- Achieved test HOTA scores of **72.32 on BFT**, **62.91 on WOSDETC**, and **65.77 on BSB**.

Identity Association for Multi-Object Tracking *Python, PyTorch*

- Developed Transformer-based identity association using appearance, geometry, and motion cues; evaluated on DanceTrack, SportsMOT, BFT, and BEE24.

Robotics and Control Projects

Wheeled-Bipedal Robot Modeling and Control
KAIST | MATLAB/Simulink, CasADi, C++, Gazebo Sep. 2024 – Oct. 2025

- Modeled the robot as a wheeled inverted pendulum and formulated velocity-tracking optimization with soft Lyapunov constraints and state and motor-torque limits.
- Built a C++ interface between Gazebo and Simulink to exchange joint states, IMU measurements, and torque commands for controller development in simulation.

Risk-Sensitive Stochastic Control (RS-JiLQG)
KAIST graduate course project | Python, NumPy Sep. 2024 – Oct. 2025

- Co-developed a risk-sensitive iterative LQG formulation to control nonlinear systems under Gaussian noise and sudden wheel-slip disturbances.
- Implemented the solver and evaluated risk settings through Monte Carlo simulations of a wheeled inverted pendulum, comparing mean and entropic costs.

PX4 Offboard Position Control with ROS 2
Python, ROS 2, PX4 Nov. 2024

- Implemented a ROS 2 node that sends position commands to PX4 and monitors vehicle state and command acknowledgments.

Additional Experience

Engineering Intern, Viettel Aerospace Institute
Embedded systems software Sep. 2023 – Feb. 2024

- Designed user interfaces for embedded systems; worked with C++, MATLAB, and Simulink.

Undergraduate Researcher, HUST
Reinforcement learning and control 2022 – 2024

- Investigated output-feedback adaptive dynamic programming (ADP) and reinforcement learning control for a two-wheeled inverted pendulum in the undergraduate thesis.
- Studied ADP-based control for mobile robots with wheel slip and wind turbines with disturbance observers; designed a PID controller for a resistance furnace.

Technical Skills

Programming: C, C++, Python | **Deep learning:** PyTorch.
Modeling and simulation: MATLAB, Simulink.
Robotics and control: Robot dynamics, optimal control, and state estimation.
Reinforcement learning: Strong foundational knowledge.
Languages: Vietnamese (native), English.

Scholarships and Awards

- **KAIST Scholarship**, M.S. program.
- **Best Graduation Thesis**, HUST thesis defense council, Feb. 2024.
- **Academic excellence scholarships**, HUST, six semesters (2020–2023).
- **Corporate scholarship for outstanding students**, 2023.

Additional Programming Projects

C++ Programming Exercises
C++, Git Apr. 2025
Maintained introductory C++ examples for language practice and version control.

Pattern Recognition with RBM Features and Neural Classification
Python, NumPy, TensorFlow | KAIST coursework Dec. 2024
Implemented a restricted Boltzmann machine feature extractor and neural classifier for MNIST, with sampling, normalization, and separate training, validation, and test sets.